

Magnetoresistance in three-dimensional systems

10

10.1 Introduction

Magnetoresistance is a general term for the changes in the components of the resistivity and conductivity tensors of materials caused by the application of magnetic field. We are going to treat the magnetoresistance of metals in a quite general and simple manner.¹ First, however, the Hall effect in a system with more than one type of carrier will be described, as it helps to illuminate the more general discussion of metals that will follow, and gives a clue as to the origins of magnetoresistance.

10.2 Hall effect with more than one type of carrier

10.2.1 General considerations

We consider the Hall effect with two or more carrier types present (e.g. electrons and holes). The geometry of a Hall effect measurement is shown in Fig. 10.1; the magnetic field $\mathbf{B}$ is applied parallel to the z direction (i.e. $\mathbf{B} = (0, 0, B)$), whilst the current I is driven through the sample in the x direction. The electric field $\mathbf{E}$ is assumed to be $\mathbf{E} = (E_x, E_y, 0)$; we assume that any effect is going to occur in the plane perpendicular to $\mathbf{B}$ because of the nature of the Lorentz force (see eqn 10.1 below). Voltage measuring contacts are provided on the sample so that E_x and E_y can be deduced (see Fig. 10.1).

We assume that the drift velocity $\mathbf{v}$ of each species of carrier can be treated using the relaxation-time approximation, i.e.

$$m^* \left\{ \frac{d\mathbf{v}}{dt} + \frac{\mathbf{v}}{\tau} \right\} = q\mathbf{E} + q\mathbf{v} \times \mathbf{B}, \quad (10.1)$$

where q is the charge of the carrier, m^* is its effective mass (assumed isotropic and energy-independent) and τ^{-1} is its relaxation (scattering) rate. Note that all changes in $\mathbf{v}$ occur in the plane perpendicular to $\mathbf{B}$; therefore it is sufficient to split eqn 10.1 into x and y components to give

$$m^* \left\{ \frac{dv_x}{dt} + \frac{v_x}{\tau} \right\} = qE_x + qv_y B \quad (10.2)$$

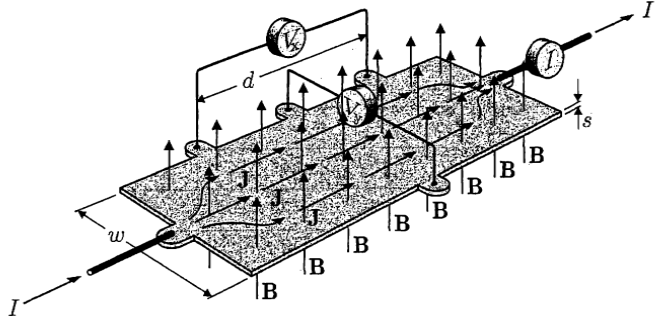
and

$$m^* \left\{ \frac{dv_y}{dt} + \frac{v_y}{\tau} \right\} = qE_y - qv_x B. \quad (10.3)$$

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¹In this chapter I shall concentrate on the magnetoresistance which arises due to modification of carrier motion due to the Lorentz force. Magnetoresistance can arise in magnetic conductors for quite different reasons, as has been already mentioned in Section 7.5.

Fig. 10.1 Geometry of a Hall effect measurement on a sample of thickness s and width w . On entering the sample, the current I becomes a current density $\mathbf{J}$ of average magnitude I/ws . The magnetic field (flux density) is uniform within the sample. The positions of voltmeters for measuring $E_x = V_x/d$ and $E_y = V_y/w$ are shown symbolically.



The Hall effect represents a steady state of the system, i.e. $dv_x/dt = dv_y/dt = 0$. Substituting eqn 10.2 into eqn 10.3 with $dv_x/dt = dv_y/dt = 0$ gives

$$\frac{m^* v_y}{\tau} = q E_y - \frac{q B \tau}{m^*} \{q E_x + q v_y B\}, \quad (10.4)$$

which can be rearranged to give

$$v_y \left\{ \frac{m^*}{\tau} + \frac{q^2 B^2 \tau}{m^*} \right\} = q E_y - \frac{q^2 B \tau}{m^*} E_x. \quad (10.5)$$

Dividing through by m^*/τ and making the identification $eB/m^* \equiv \omega_c$ (i.e. the cyclotron frequency) gives

$$v_y \{1 + \omega_c^2 \tau^2\} = \frac{q \tau}{m^*} \{E_y - \omega_c \tau E_x\}. \quad (10.6)$$

Hall effect experiments are usually quite deliberately carried out at low magnetic fields, such that $\omega_c \tau \ll 1$, implying that terms $\sim \omega_c^2 \tau^2$ can be neglected. Therefore eqn 10.6 becomes

$$v_y = \frac{q \tau}{m^*} \{E_y - \omega_c \tau E_x\} = \frac{q \tau}{m^*} \{E_y - \frac{q B \tau}{m^*} E_x\}. \quad (10.7)$$

We now consider an arbitrary number of carrier types, with each type being labelled by the integer j ; the j th carrier type has effective mass m_j^* , charge q_j , scattering rate τ_j and number density n_j . For each carrier type, eqn 10.7 therefore becomes

$$v_{y,j} = \frac{q_j \tau_j}{m_j^*} \{E_y - \frac{q_j B \tau_j}{m_j^*} E_x\}. \quad (10.8)$$

Now the net transverse current must be zero, as there is nowhere for it to go (see Fig. 10.1). Therefore

$$\sum_j n_j v_{y,j} q_j = 0. \quad (10.9)$$

Equations 10.8 and 10.9 can be used to derive the Hall coefficient for an arbitrary number of carrier types. We shall use them to treat the simple case of electrons and heavy holes in a semiconductor; as usual, the light holes, with their relatively feeble density of states compared to that of the heavy holes, will be ignored.

10.2.2 Hall effect in the presence of electrons and holes

In the case of electrons in the conduction band (with effective mass m_c^* , scattering rate τ_c^{-1} , charge q_c , density n) and heavy holes in the valence band (with effective mass m_{hh}^* , scattering rate τ_{hh}^{-1} , charge q_{hh} , density p), eqns 10.8 and 10.9 combine to give

$$\frac{nq_c^2\tau_c}{m_c^*}\{E_y - \frac{q_c B \tau_c}{m_c^*}E_x\} + \frac{pq_{hh}^2\tau_{hh}}{m_{hh}^*}\{E_y - \frac{q_{hh} B \tau_{hh}}{m_{hh}^*}E_x\} = 0. \quad (10.10)$$

Equation 10.10 can be rearranged to give

$$E_y\{n\mu_c + p\mu_{hh}\} = E_x\{p\mu_{hh}^2 - n\mu_c^2\}B, \quad (10.11)$$

where $\mu_{hh} = |q_{hh}\tau_{hh}/m_{hh}^*|$ is the heavy hole mobility and where $\mu_c = |q_c\tau_c/m_c^*|$ is the electron mobility, and I have substituted $q_{hh} \equiv +e$ and $q_c \equiv -e$. Now

$$E_x = \frac{J_x}{\sigma} = \frac{J_x}{|e|(n\mu_c + p\mu_{hh})}, \quad (10.12)$$

where J_x is the current density in the x direction. Combining eqns 10.11 and 10.12 gives

$$R_H \equiv \frac{E_y}{J_x B} = \frac{1}{|e|} \frac{(p\mu_{hh}^2 - n\mu_c^2)}{(n\mu_c + p\mu_{hh})^2}. \quad (10.13)$$

This treatment is explored in more depth in Exercises 10.1 and 10.2.²

²Some excellent illustrative data are shown in Fig. 4.3 of *Semiconductor physics*, by K. Seeger (Springer, Berlin, 1991).

10.2.3 A clue about the origins of magnetoresistance

Equations 10.9 and 10.10 show that, although no *net* current flows in the y direction, the currents carried in the y -direction by a particular type of carrier may (i.e. probably will) be non-zero. Carriers flowing in the y -direction will experience a Lorentz force caused by $\mathbf{B}$ in the *negative x -direction* (you can satisfy yourself that this will always be the case). This backflow of carriers will act to change the apparent resistivity E_x/J_x , i.e. cause magnetoresistance.³

In the following section we shall explore this idea more formally and in a very general manner. We shall see that the presence of more than one 'carrier type' (and here the term is used very imprecisely) is necessary for magnetoresistance to be observed.

In order to treat all of the different contributions to the conductivity and resistivity in a sanitary fashion, we shall introduce the idea of conductivity and resistivity tensors.

³Those who are unconvinced by this hand-waving argument should go back to eqn 10.6 and repeat the above derivation for $\omega_c\tau \gg 1$. They will find that $((p/\mu_{hh}) + (n/\mu_c))E_y = (p - n)E_x B$ (i.e. the Hall field is zero if $n = p$). Putting $n = p$ yields $J_x = ((p/\mu_{hh}) + (n/\mu_c))eE_x/B^2$, i.e. $\rho \propto B^2$. This is the reason for the very large magnetoresistance in compensated semimetals (equal number of holes and electrons at Fermi surface) such as Bi.

10.3 Magnetoresistance in metals

10.3.1 The absence of magnetoresistance in the Sommerfeld model of metals

We consider first of all a metal with a simple spherical Fermi surface and isotropic, energy-independent effective mass. As in Section 10.2.1, the magnetic field $\mathbf{B}$ will be parallel to z (see Fig. 10.1), and we shall use the same

symbols (effective mass m^* , scattering rate τ^{-1} and electronic charge $-e$). Applying eqn 10.1, we have

$$m^* \left\{ \frac{d\mathbf{v}}{dt} + \frac{\mathbf{v}}{\tau} \right\} = -e\mathbf{E} - e\mathbf{v} \times \mathbf{B}. \quad (10.14)$$

Two things may be deduced from this equation.

- The motion of the electrons in the direction parallel to $\mathbf{B}$ is unaffected. Therefore there will be no longitudinal magnetoresistance; in this context, the longitudinal resistivity is measured in the direction parallel to the field $\mathbf{B}$ (i.e. both the applied current density and the measured electric field are parallel to $\mathbf{B}$).
- There may well be transverse magnetoresistance. Here *transverse* resistivity means that measured in the direction perpendicular to the field $\mathbf{B}$ (i.e. both the applied current density and the measured electric field are in the plane perpendicular to $\mathbf{B}$).

Let us look at the second point in more detail. To simplify matters, we shall initially consider an electric field directed only along the x direction (i.e. $\mathbf{E} = (E_x, 0, 0)$); our tactic will be to deduce the components of the current density $\mathbf{J} = (J_x, J_y, 0)$ that flow in response to $\mathbf{B}$ and $\mathbf{E}$.

As in the previous section, we are dealing with a steady state of the system, i.e. $dv_x/dt = dv_y/dt = 0$. Taking $\mathbf{B} = (0, 0, B)$ and $\mathbf{E} = (E_x, 0, 0)$ as defined above, we rewrite eqn 10.2 and eqn 10.3 in the form

$$v_{d,x} = -\frac{e\tau}{m^*} \{E_x + v_{d,y}B\} \quad (10.15)$$

and

$$v_{d,y} = \frac{e\tau}{m^*} v_{d,x}B, \quad (10.16)$$

where the subscript 'd' emphasises the fact that we are dealing with a drift velocity.

Equations 10.15 and 10.16 show that the magnetic field has made the conductivity anisotropic; it has become a tensor, rather than a scalar. In order to work out the components of the conductivity tensor, we look at the current densities $J_x = -nev_{d,x}$ and $J_y = -nev_{d,y}$. Substituting these into eqns 10.15 and 10.16 yields, after some rearrangement (see Exercise 3)

$$J_x = \sigma_{xx}E_x \text{ and } J_y = \sigma_{yx}E_x,$$

where

$$\sigma_{xx} = \frac{\sigma_0}{1 + \omega_c^2\tau^2} \quad (10.17)$$

and

$$\sigma_{yx} = \frac{\sigma_0\omega_c\tau}{1 + \omega_c^2\tau^2}. \quad (10.18)$$

Here $\sigma_0 = ne^2\tau/m^*$ is the zero-field conductivity in the Sommerfeld model and $\omega_c = eB/m^*$ is the cyclotron frequency.

The conductivity tensor shows that, in a magnetic field, the total current density $\mathbf{J}$ no longer flows parallel to the applied E -field, E_x ; instead, it



Fig. 10.2 Geometrical interpretation of the components of current density J_x , J_y and caused by electric field component E_x and magnetic field $(0, 0, B)$; $\mathbf{J}$ is the total current density.

now contains both x and y components. Figure 10.2 gives a geometrical interpretation of $\mathbf{J}$ and the components of current density J_x , J_y caused by electric field component E_x and magnetic field $(0, 0, B)$.

Equation 10.17 shows that as $B \rightarrow \infty$, $\sigma_{xx} \propto B^{-2}$. We might therefore expect to see some magnetoresistance. However, most experiments (see Fig. 10.1) measure *voltages* dropped in the x - and y -directions between pairs of contacts, rather than measuring the x and y components of the current density. In such experiments the current is forced to go along the x direction, so that $\mathbf{J} \equiv J_x \mathbf{e}_1$; in contrast, the electric field will have components in both x and y directions (see Fig. 10.3). Therefore we want the components

$$\rho_{xx} \equiv \frac{E_x}{J_x} \text{ and } \rho_{yx} \equiv \frac{E_y}{J_x} \quad (10.19)$$

of the *resistivity* tensor, rather than the conductivity.

The general conductivity tensor is

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix}. \quad (10.20)$$

The derivations which start at eqns 10.15 and 10.16 can be repeated with $\mathbf{E} = (0, E_y, 0)$ to yield $\sigma_{xy} = -\sigma_{yx}$ and $\sigma_{yy} = \sigma_{xx}$ (see Exercise 10.3),⁴ so that we have

$$\sigma = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ -\sigma_{xy} & \sigma_{xx} \end{pmatrix} = \frac{\sigma_0}{1 + \omega_c^2 \tau^2} \begin{pmatrix} 1 & -\omega_c \tau \\ \omega_c \tau & 1 \end{pmatrix}. \quad (10.21)$$

This tensor can then be inverted using standard methods to give the resistivity tensor

$$\rho = \begin{pmatrix} \rho_{xx} & \rho_{xy} \\ \rho_{yx} & \rho_{yy} \end{pmatrix} = \frac{1}{\sigma_0} \begin{pmatrix} 1 & \omega_c \tau \\ -\omega_c \tau & 1 \end{pmatrix}. \quad (10.22)$$

The components of interest in the experimental arrangement shown in Fig. 10.1 are

$$\rho_{xx} = \rho_0 \text{ and } \rho_{yx} = -\rho_0 \omega_c \tau = -\frac{B}{ne}, \quad (10.23)$$

where $\rho_0 = 1/\sigma_0$ (see Fig. 10.3). Therefore we get no magnetoresistance in the diagonal components of the resistivity tensor and the familiar Hall effect for one carrier in the off-diagonal components.

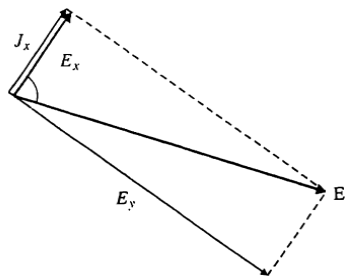


Fig. 10.3 Geometrical interpretation of the components of electric field E_x , E_y and the total field $\mathbf{E}$ caused by current density component J_x and magnetic field $(0, 0, B)$.

⁴This fact can also be deduced using symmetry considerations.

10.3.2 The presence of magnetoresistance in real metals

Almost all real metals exhibit some form of magnetoresistance, and so we must try to find out what is wrong with the approach above. In the above derivation

we assumed that all carriers had the same value of m^* and τ . However, in a real metal we could have

- electrons with different values of m^* (e.g. from anisotropic bands);
- electrons with different values of τ (e.g. some parts of the Fermi surface may have higher scattering probabilities than others);
- a combination of both.

We therefore split the current density $\mathbf{J}$ into several components

$$\mathbf{J}_j = \sigma_{xx,j} E_x \mathbf{e}_1 + \sigma_{yx,j} E_y \mathbf{e}_2 \quad (10.24)$$

where the index j indicates a contribution from the j th type of carrier. Each type of carrier will have a different density n_j and/or effective mass m_j^* and/or scattering rate τ_j^{-1} . Hence, the components of the conductivity tensor $\sigma_{xx,j}$ and $\sigma_{yx,j}$ will differ for each type of carrier, resulting in $\mathbf{J}_j$ s which do not in general point in the same direction as each other.

The total current $\mathbf{J}$ is just the sum of all of the components

$$\mathbf{J} = \sum_j \mathbf{J}_j. \quad (10.25)$$

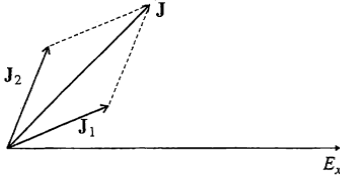


Fig. 10.4 Geometrical interpretation of the components of current density $\mathbf{J}_1$ and $\mathbf{J}_2$ due to two different species of carrier caused by electric field component E_x and magnetic field $(0, 0, B)$.

In order to see what happens, we take a very simple case of just two carrier types, $j = 1, 2$. Equation 10.24 shows that, barring some very unlikely coincidence, $\mathbf{J}_1$ and $\mathbf{J}_2$ will be in different directions. This situation is illustrated in Fig. 10.4; the application of the magnetic field means that $\mathbf{J}_1$ and $\mathbf{J}_2$ are no longer parallel, so that

$$|\mathbf{J}| \leq |\mathbf{J}_1 + \mathbf{J}_2|, \quad (10.26)$$

i.e. the resistivity increases with increasing magnetic field. We therefore have magnetoresistance.

We note in passing that, as above, $\sigma_{xx} \propto B^{-2}$ as $B \rightarrow \infty$ (see eqn 10.17 and the paragraph following it). This will be important in the discussion of the following section.

10.3.3 The use of magnetoresistance in finding the Fermi-surface shape

We consider first a *closed* section of Fermi surface, about which a carrier can perform closed orbits under the influence of a magnetic field (see Fig. 10.5). As $B \rightarrow \infty$, $\omega_c \tau \rightarrow \infty$, so that an electron will tend to make many circuits of the Fermi surface before scattering. Therefore, the velocity of the electron in the plane perpendicular to $\mathbf{B}$ will average to zero; this is the reason why σ_{xx} and σ_{yy} both vary as B^{-2} in very high fields. Using the conductivity tensor components, the current densities can be written

$$J_x = \sigma_{xx} E_x + \frac{1}{RB} E_y \quad (10.27)$$

and

$$J_y = -\frac{1}{RB} E_x + \sigma_{yy} E_y \quad (10.28)$$

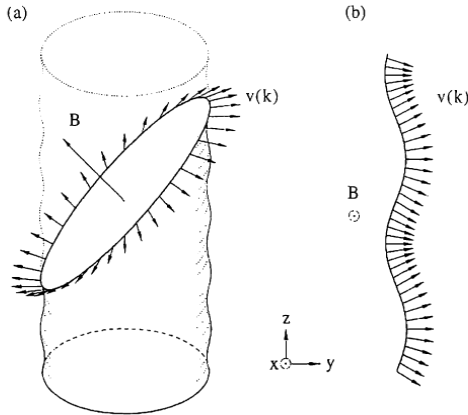


Fig. 10.5 (a) Schematic of electron motion on a closed section of Fermi surface in a magnetic field. The arrows indicate the velocities of an electron following a closed orbit about the Fermi surface in a plane perpendicular to the magnetic field B . (b) An open orbit on the Fermi surface. In an in-plane magnetic field, electrons will be driven across the Fermi surface, so that their velocities (shown by arrows) will rock from side to side.

where R is the Hall coefficient, and the off-diagonal tensor components have been written $\sigma_{xy} = 1/RB$ and $\sigma_{yx} = -1/RB$. Eliminating E_y gives

$$E_x = \frac{1}{\sigma_{xx} + (R^2 B^2 \sigma_{yy})^{-1}} J_x + \frac{RB}{1 + R^2 B^2 \sigma_{xx} \sigma_{yy}} J_y. \quad (10.29)$$

As mentioned above, σ_{xx} and σ_{yy} both vary as B^{-2} in very high fields, so that as $B \rightarrow \infty$, $(B^2 R^2 \sigma_{yy})^{-1} \gg \sigma_{xx}$. Therefore, $\rho_{xx} = E_x/J_x$ tends to a constant at high fields, i.e. it saturates.

We now turn to an *open* section of Fermi surface (see Fig. 10.5) about which an electron cannot perform closed orbits under the influence of a magnetic field. In this case, even as $B \rightarrow \infty$, the average value of v_y remains finite; therefore $\sigma_{yy} \rightarrow C$, a constant. Substituting $\sigma_{yy} = C$ and $\sigma_{xx} = \mathcal{A}B^{-2}$, where $\mathcal{A}$ is another constant, into eqn 10.29 yields

$$\rho_{xx} = \frac{E_x}{J_x} = \frac{1}{\mathcal{A}B^{-2} + (R^2 B^2 C)^{-1}} \propto B^2, \quad (10.30)$$

i.e. $\rho_{xx} \propto B^2$ as $B \rightarrow \infty$.

We therefore have two distinct results

- **closed orbits** produce a ρ_{xx} which saturates as $B \rightarrow \infty$;
- **open orbits** produce a ρ_{xx} proportional to B^2 as $B \rightarrow \infty$.

This has been used to great effect in elucidating the Fermi surface of metals such as copper, where changing the orientation of the magnetic field can produce open or closed orbits about the Fermi surface (see Exercise 10.4).

10.4 The magnetophonon effect

Oscillations can be observed in the resistivity of both bulk and two-dimensional semiconductors at elevated temperatures ~ 100 K; this is known as the magnetophonon effect or magnetophonon resonance. The effect

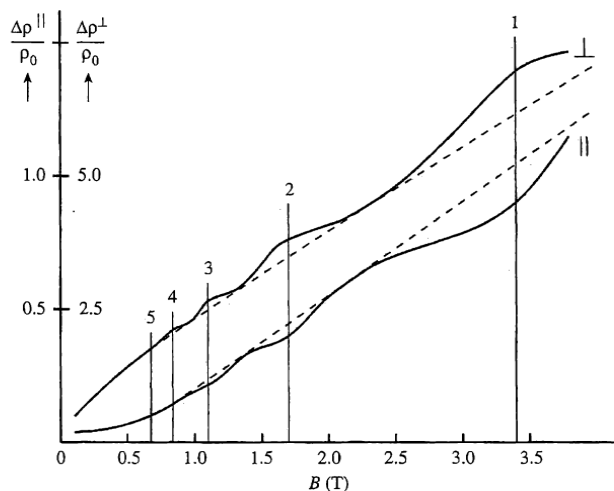


Fig. 10.6 Magnetophonon resonances in the longitudinal and transverse resistivities of InSb at 90 K.

is caused by resonant inter-Landau-level scattering of electrons by long-wavelength longitudinal optic (LO) phonons; such phonons are very effective scatterers of electrons. (Why? think about the type of polarisation field that they produce; see Section D.2.) As such phonons have virtually zero wavevector, the transition is 'vertical', i.e. between almost identical points in k -space in the initial and final Landau levels involved. By conservation of energy, the condition for the magnetophonon effect to occur is therefore

$$j\omega_c = \omega_{LO}, \quad (10.31)$$

where ω_{LO} is the phonon frequency, i.e. an integer number j of Landau level spacings matches an LO phonon energy. This leads to oscillations in the resistivity periodic in $1/B$; if the phonon frequency is known, the effective mass can be deduced from eqn 10.31.

The conditions for magnetophonon resonance to be observed are

- the temperature should be low enough for the Landau levels to be resolved;
- the temperature should be high enough for a substantial population of LO phonons.

In practice, 70–100 K seems to be a good compromise. Figure 10.6 shows magnetophonon resonances in InSb at 90 K.

Further reading

Some useful general reading on magnetoresistance is contained in *Electrons in metals and semiconductors*, by R.G. Chambers (Chapman and Hall, London, 1990) Chapters 1, 2 and 11, *Semiconductor physics*, by K. Seeger (Springer, Berlin, 1991) Chapter 9 (hard), *Solid state physics*, by N.W. Ashcroft and N.D.

Mermin (Holt, Rinehart and Winston, New York, 1976) Chapters 12, 13 and 15 (hard).

Those who wish to calculate magnetoresistance, or who are interested in seeing how the effect helped to determine the Fermi surfaces of a variety of metals are referred to *Magnetoresistance in metals* by A.B. Pippard (Cambridge University Press, 1989).

The angle dependence of the magnetoresistance is of great importance in discovering the Fermi-surface topologies of new materials such as the organic conductors and oxides described in Chapter 7. Recent reviews (including suggestions for tractable calculations based on the Boltzmann transport equation and reasonably up-to-date literature) are given by S.J. Blundell *et al.* (*Phys. Rev. B* **53**, 5609 (1996); *Journal de Physique I*, **6**, 1837 (1996)), A. Ardavan *et al.* (the inclusion of high-frequency effects; *Phys. Rev. B* **60**, 15500 (1999)) and E. Ohmichi *et al.* (application to Sr_2RuO_4 ; *Phys. Rev. B* **59**, 7263 (1999)).

A good review of the magnetophonon effect with relatively recent data is given in *Landau level spectroscopy*, edited by G. Landwehr and E.I. Rashba, Volume 27 of *Modern problems in condensed matter sciences*, edited by V.M. Agranovich and A.A. Maradudin (Elsevier Science Publishers, Amsterdam 1991), Chapters 13 and 20.

Exercises

- (10.1) Explain why the electron mobility μ_e is generally much greater than the hole mobility μ_h in many direct-gap semiconductors such as InSb and GaAs.

Figure 10.7 shows R_H plotted as a function of the inverse temperature for a sample of InSb. R_H is observed to change sign between 0.005 and 0.006 K^{-1} . Describe the mechanisms responsible for the form of the curve and estimate (i) the concentration and type of impurities in the sample and (ii) the band-gap of InSb. What temperature dependence would be expected for the mobilities of the carriers over the temperature range shown?

- (10.2)* Show that the Hall coefficient R_H of a material containing two types of carrier, with respective concentrations n_1 and n_2 , charges q_1 and q_2 and mobilities μ_1 and μ_2 can be expressed as

$$R_H = \frac{1}{e^2} \frac{n_1 q_1 \mu_1^2 + n_2 q_2 \mu_2^2}{(n_1 \mu_1 + n_2 \mu_2)^2}, \quad (10.32)$$

where e is the electronic charge, indicating the approximations inherent in your derivation.

When GaAs is subjected to hydrostatic pressure, the band-gap changes from direct to indirect at a pressure of about 40 kbar, due to the reordering of the conduction band minima at the zone centre and near certain zone boundaries. Describe briefly why the application of pressure affects the shape and width of the bands.

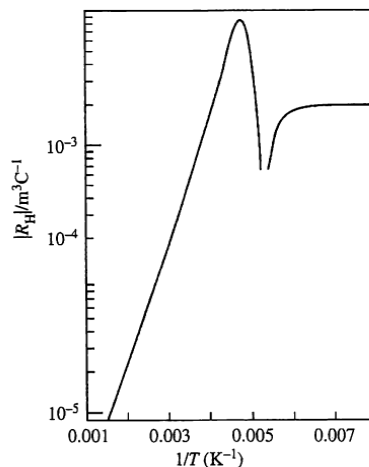


Fig. 10.7 Hall coefficient of an InSb sample versus $1/T$.

Room temperature measurements of (a) the Hall coefficient, normalised to its ambient pressure value, and (b) the Hall mobility, are plotted for n -type GaAs as a function of pressure in Fig. 10.8. What do the data tell you about the conduction band minima involved?

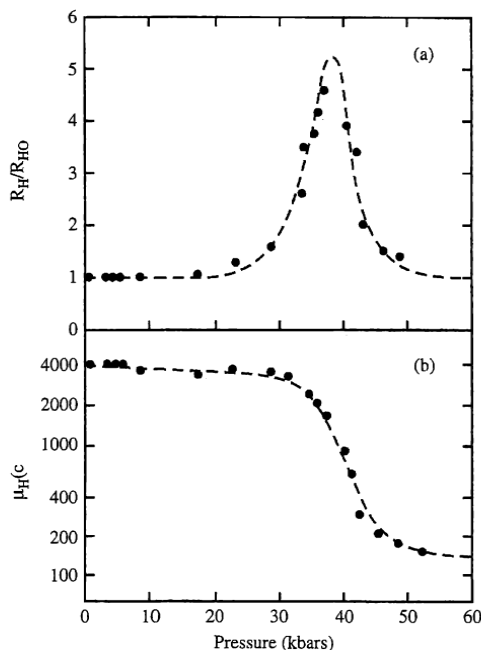


Fig. 10.8 Room temperature measurements of (a) the Hall coefficient, normalised to its ambient pressure value, and (b) the Hall mobility for *n*-type GaAs as a function of pressure.

Use the above formula to show that the peak in the normalised Hall coefficient occurs when $ab \approx 1$, where $a = n_2/n_1$ and $b = \mu_2/\mu_1 \ll 1$. Show that the height of the peak is approximately equal to $(4b)^{-1}$, and compare the value of b deduced this way to that obtained directly from the pressure dependence of the

mobility in Fig. 10.8(b). What does this comparison imply about the strength of scattering between the two types of carrier, relative to the strength of scattering between each type of carrier and the lattice?

- (10.3) Starting with eqns 10.15 and 10.16 with $\mathbf{E} = (E_x, 0, 0)$, derive the conductivity tensor components σ_{xx} and σ_{yx} . Repeat the derivation using $\mathbf{E} = (0, E_y, 0)$, to show that $\sigma_{yy} = \sigma_{xx}$ and $\sigma_{xy} = -\sigma_{yx}$. Invert the resulting conductivity tensor to find the components of the resistivity tensor.
- (10.4) Figure 10.9 shows the magnetoresistance of copper; experimental details are given in the caption. Explain qualitatively the form of the magnetoresistance. (You may find it helpful to refer to Figs. 4.8 and 8.7.)

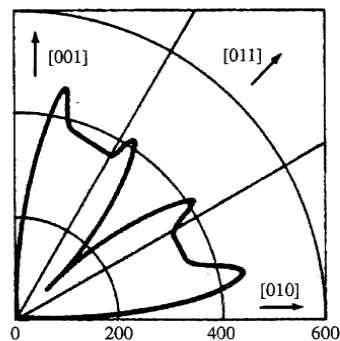


Fig. 10.9 Magnetoresistance of copper at a temperature of 4.2 K and a fixed magnetic field of 1.8 T; the current has been applied in the [100] direction (perpendicular to the plane of the page) and the magnetic field has been rotated from the [001] direction to the [010] direction. The magnetoresistance has been plotted radially as $(\rho(B) - \rho(B=0))/\rho(B=0)$. (Data from J.R. Klauder and J.E. Kunzler, *The Fermi surface*, edited by W. Harrison (Wiley, New York, 1960).)